

LAB 2: AGILE BACKLOG CREATION & SPRINT SIMULATION IN JIRA

Project: Hospital Bed & ICU Allocation Dashboard

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1. EPICS AND USER STORIES

The Lab 1 requirements were converted into **4 Epics and 7 User Stories**. The User Stories were prioritized based on their importance to the hospital system and estimated using Fibonacci-based Story Points.

Epic 1 – Patient Triage & Prioritization

Focuses on assessing patient severity and prioritizing patients based on clinical acuity.

- **User Story 1.1:** Calculate Triage Severity and Prioritize ER Waitlist
Priority: High | **Story Points:** 5

Epic 2 – Bed Management & Allocation

Focuses on managing ER/ICU bed status and allocating suitable beds to high-priority patients.

- **User Story 2.1:** View and Update ER/ICU Bed Status
Priority: High | **Story Points:** 3
- **User Story 2.2:** Allocate Available ER/ICU Bed to Highest-Priority Patient
Priority: High | **Story Points:** 5

Epic 3 – Inter-Facility Transfer Coordination

Focuses on transferring patients when a suitable ICU bed is unavailable and coordinating ambulance assignment.

- **User Story 3.1:** Initiate Inter-Facility ICU Transfer
Priority: High | **Story Points:** 5

- **User Story 3.2:** Assign Ambulance to Approved Transfer
Priority: Medium | Story Points: 3

Epic 4 – System Performance & Security

Focuses on real-time bed updates and secure, role-based access.

- **User Story 4.1:** Propagate Real-Time Bed Status Updates
Priority: High | Story Points: 8
- **User Story 4.2:** Restrict System Access by User Role
Priority: High | Story Points: 5

Total Backlog: 34 Story Points

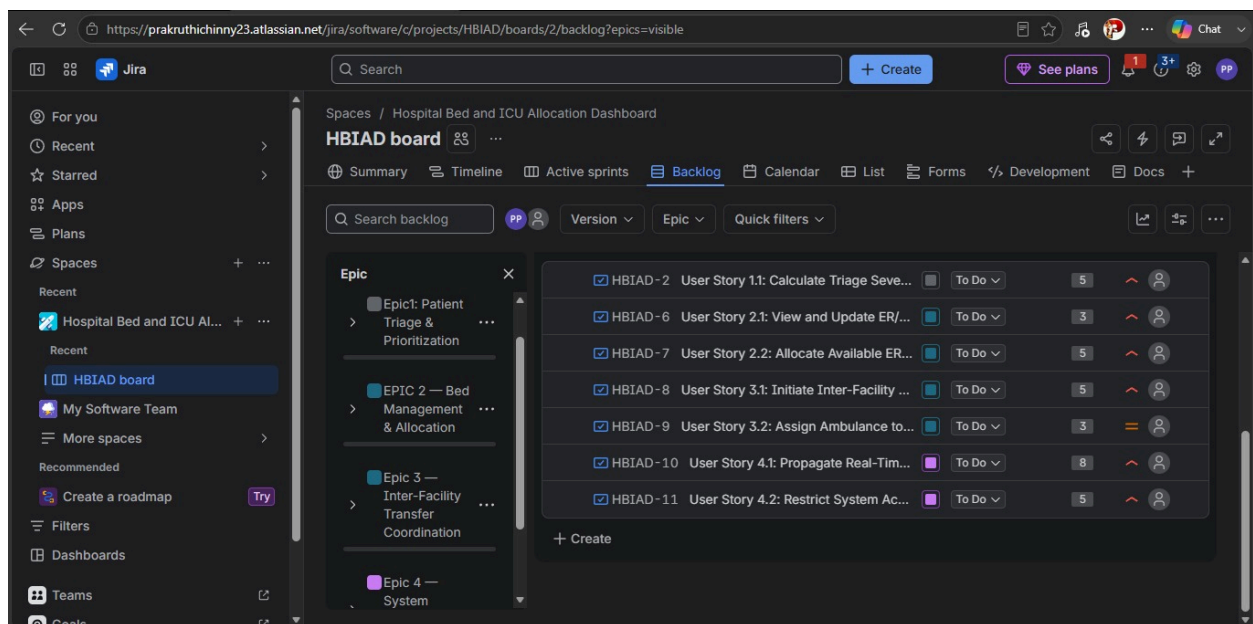


Figure 1: Jira backlog showing Epics, User Stories and priorities.

2. STORY POINT ESTIMATION

Story Points were assigned using the **Fibonacci sequence**, considering the complexity, effort, and uncertainty of each User Story.

The User Stories were estimated as **3, 5, and 8 Story Points**, with a total backlog estimate of **34 Story Points**.

3. SPRINT 1

Sprint 1 was planned as a **one-week sprint** with **21 Story Points**.

It included:

- Calculate Triage Severity and Prioritize ER Waitlist – **5 SP**
- View and Update ER/ICU Bed Status – **3 SP**
- Allocate Available ER/ICU Bed – **5 SP**
- Propagate Real-Time Bed Status Updates – **8 SP**

The selected User Stories were added to the sprint and progressed through **To Do** → **In Progress** → **Done**.

3.1 Creation of Sprint 1

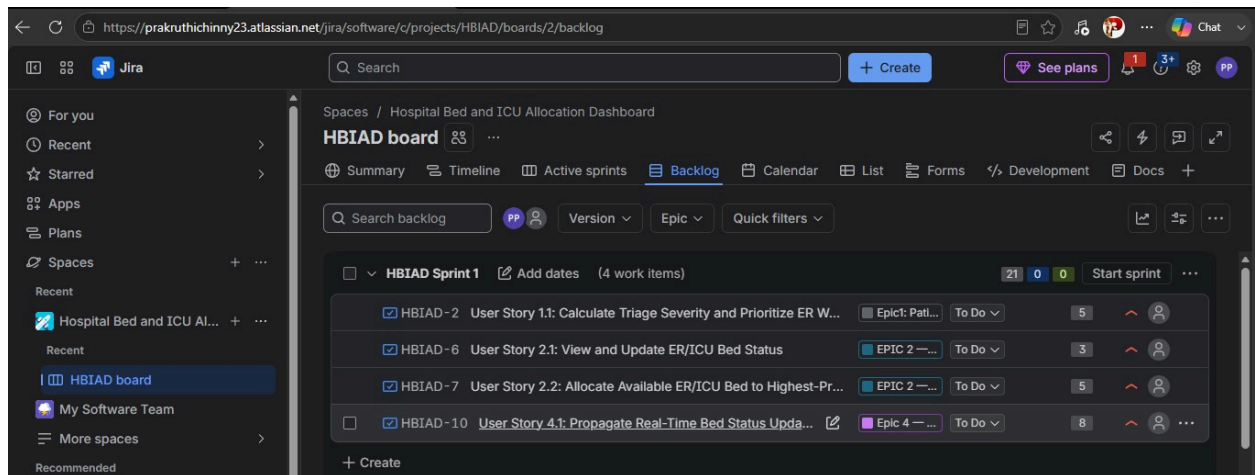


Figure 2: Sprint 1 created and planned in Jira.

3.2 Starting Sprint 1

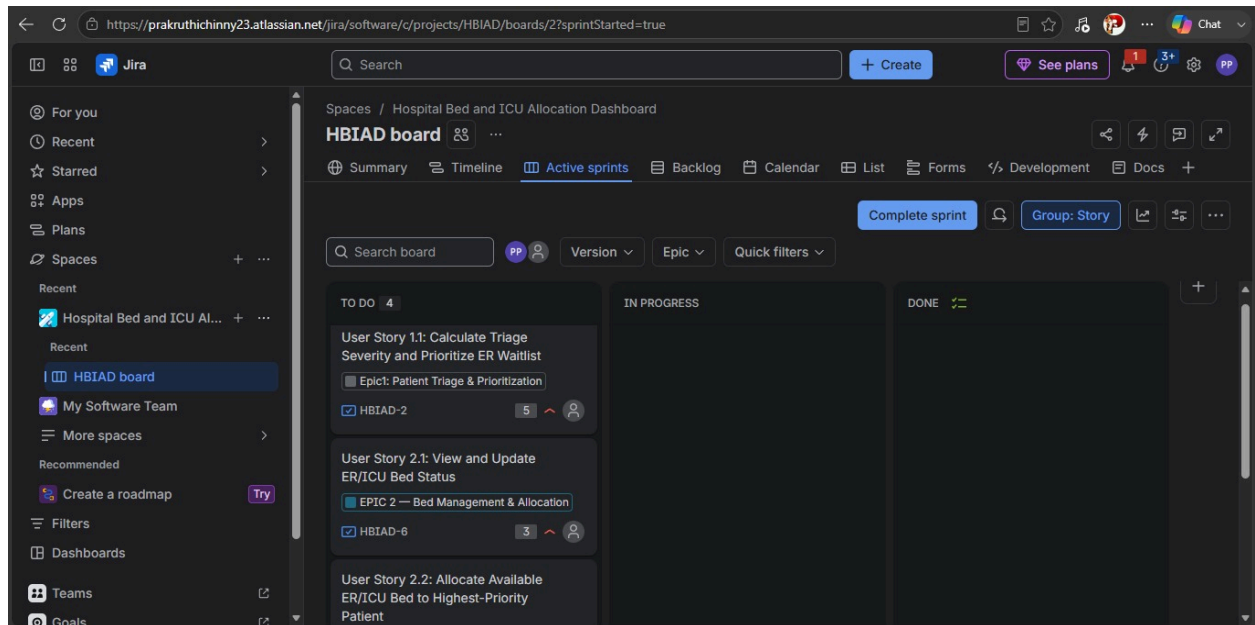


Figure 3: Sprint 1 started in Jira.

3.3 Intermediate Progress of Sprint 1

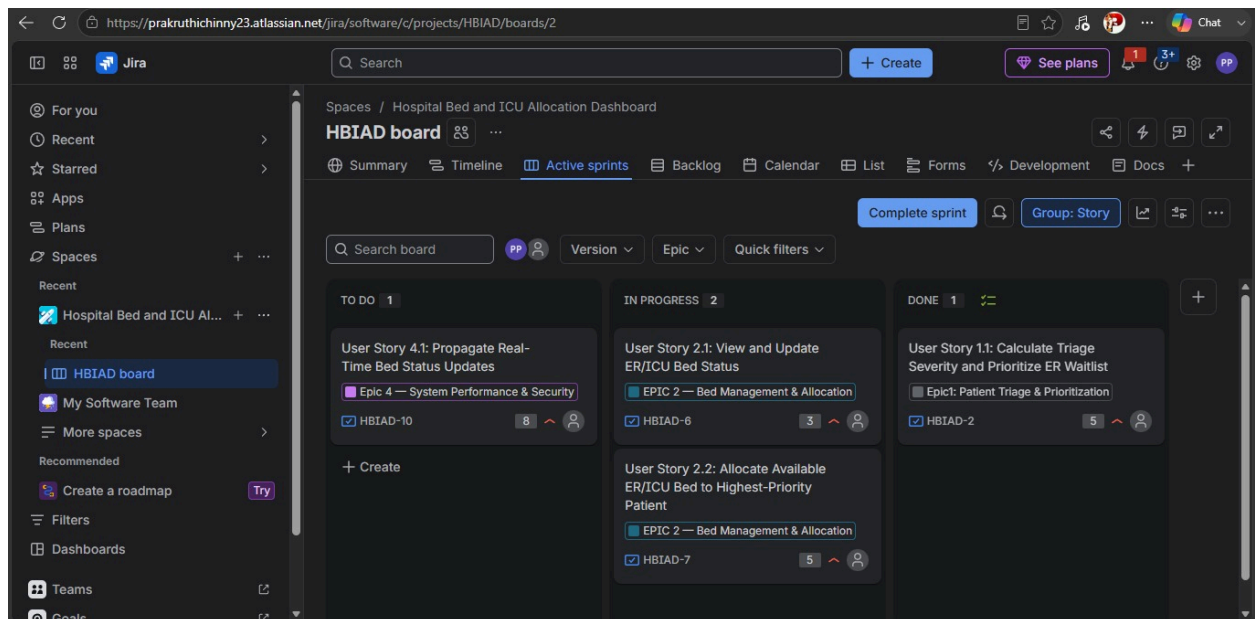


Figure 4: Intermediate progress of Sprint 1.

3.4 Completion of Sprint 1

All **21 Story Points** planned for Sprint 1 were completed.

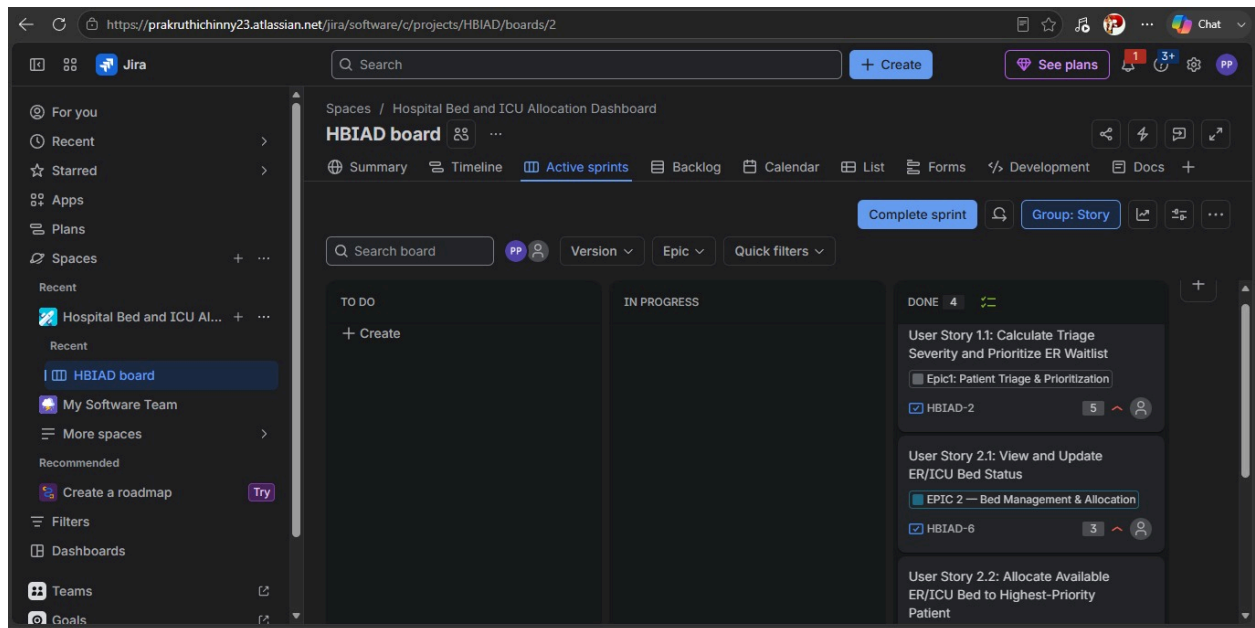


Figure 5: Sprint 1 completed successfully.

4. SPRINT 2

Sprint 2 was created using the remaining User Stories with a total of **13 Story Points**.

It included:

- Initiate Inter-Facility ICU Transfer – **5 SP**
- Assign Ambulance to Approved Transfer – **3 SP**
- Restrict System Access by User Role – **5 SP**

4.1 Creation of Sprint 2

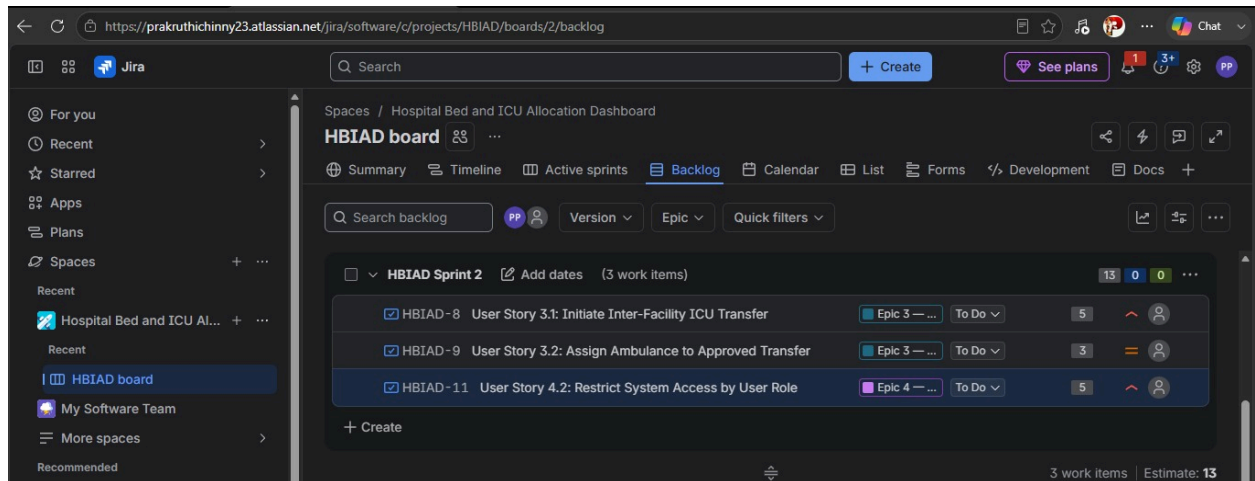


Figure 6: Sprint 2 created and planned in Jira.

4.2 Starting Sprint 2

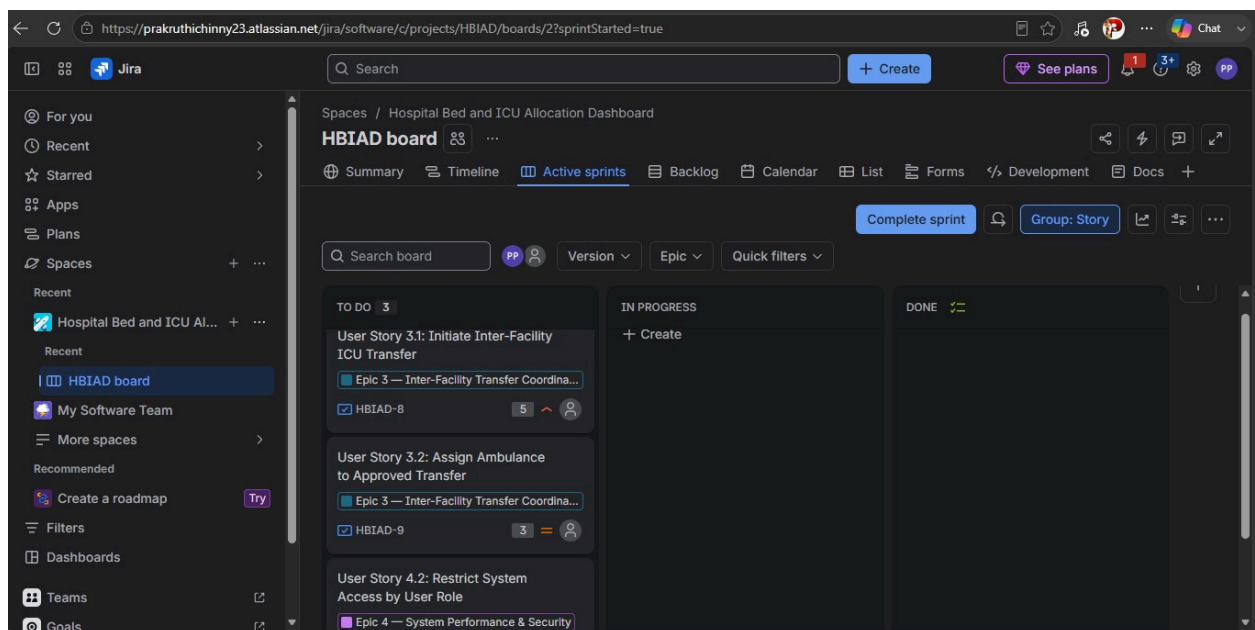


Figure 7: Sprint 2 started in Jira.

4.3 Intermediate Progress of Sprint 2

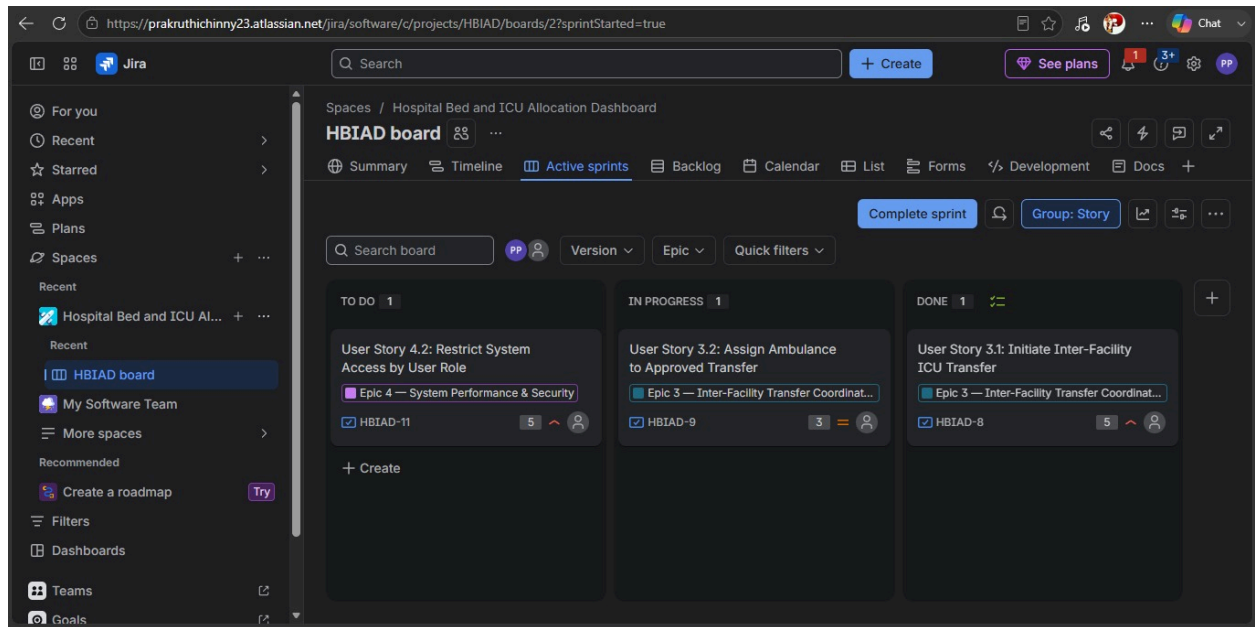


Figure 8: Intermediate progress of Sprint 2.

4.4 Completion of Sprint 2

All 13 Story Points planned for Sprint 2 were completed.

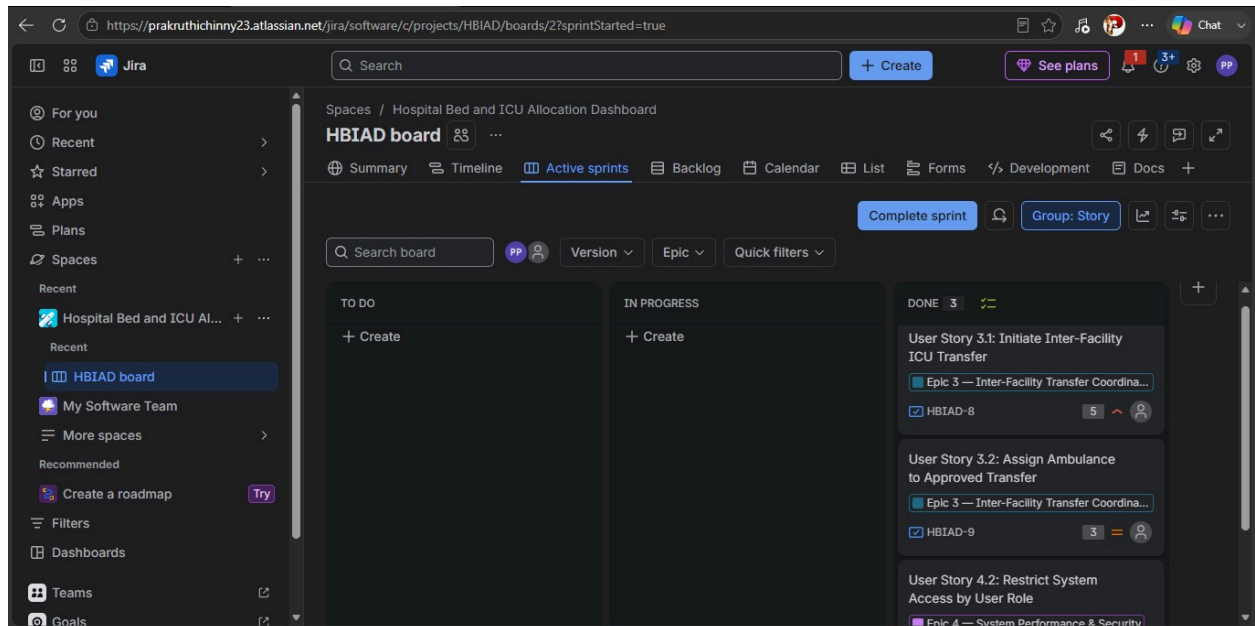


Figure 9: Sprint 2 completed successfully.

5. BURNDOWN CHART

5.1 Sprint 1 Burndown Chart

The Sprint 1 Burndown Chart shows the reduction of remaining work from **21 Story Points to 0**, indicating that all planned Sprint 1 work was completed.

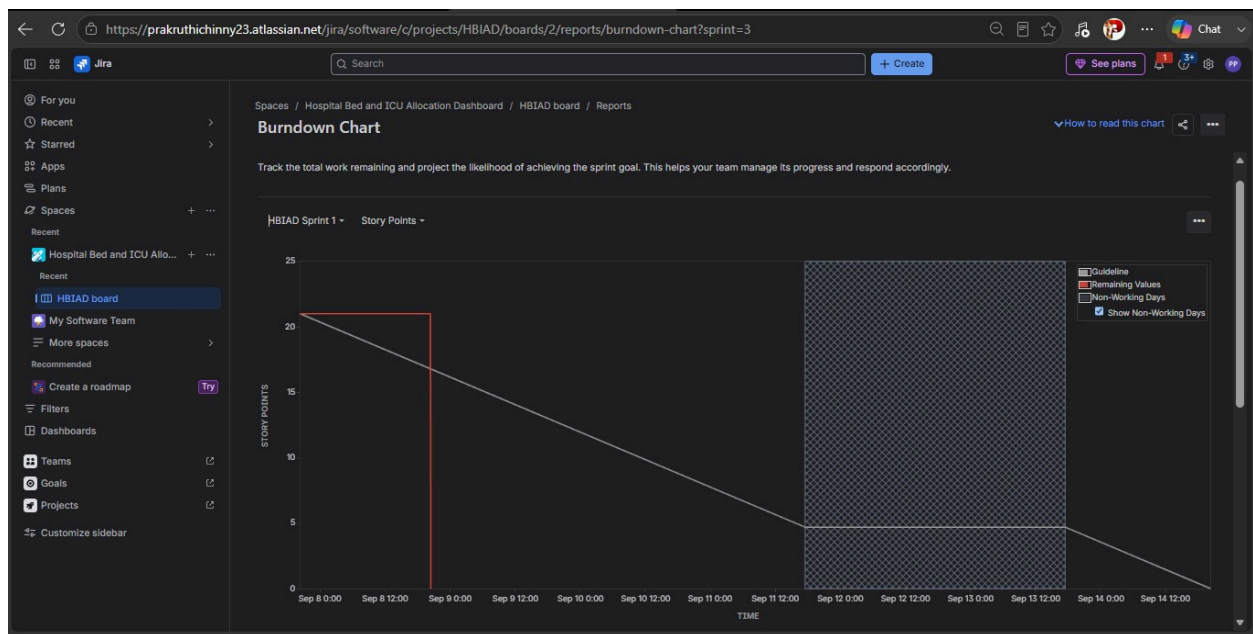


Figure 10: Sprint 1 Burndown Chart.

5.2 Sprint 2 Burndown Chart

The Sprint 2 Burndown Chart shows the reduction of remaining work from **13 Story Points to 0**, indicating successful completion of the planned Sprint 2 workload.

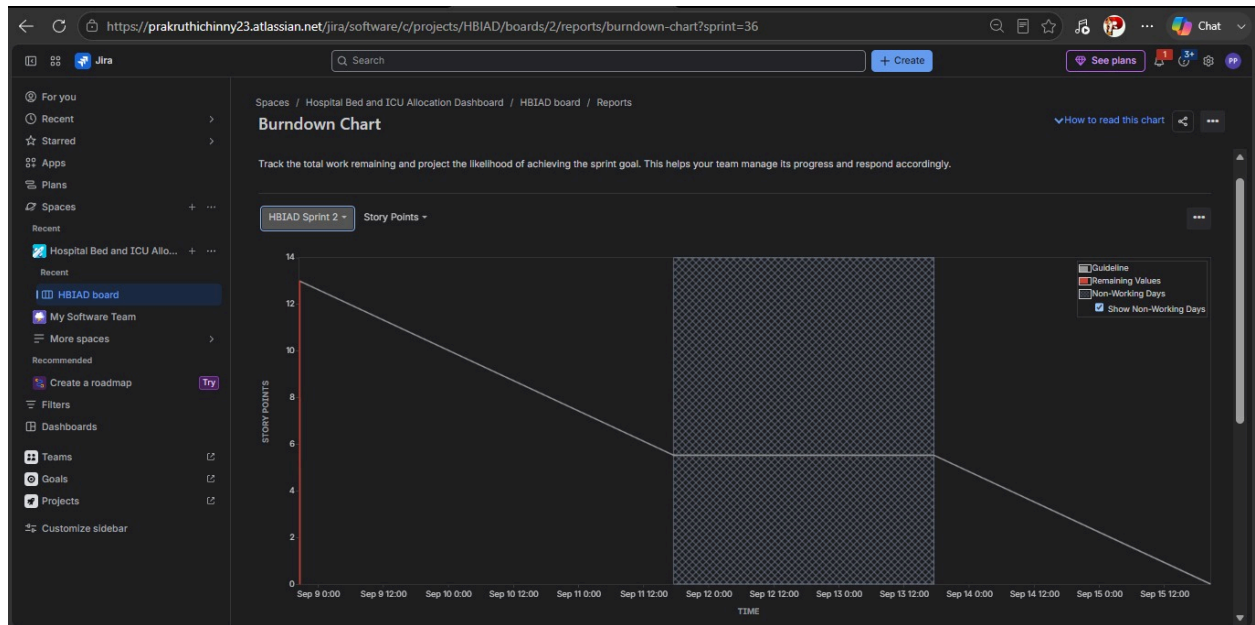


Figure 11: Sprint 2 Burndown Chart.

6. REFLECTION

6.1 Did your estimations reflect the actual effort?

Yes. The estimates were close to the actual effort. The more complex tasks were given higher Story Points, and we were able to complete both sprints as planned.

6.2 Was your backlog well-prioritized?

Yes. We gave High priority to the important features like triage, bed allocation, transfers, real-time updates, and security. Ambulance assignment was given Medium priority.

6.3 How did your simulated sprint align with your plan?

The sprints followed our plan. Sprint 1 had **21 Story Points** and Sprint 2 had **13 Story Points**, and all the planned stories were completed.

6.4 What insights did the burndown chart give about your team's capacity?

The charts showed that we were able to complete the planned work in both sprints. It also gave us an idea of how much work we could handle in a sprint.

